

SMART INDIA HACKATHON 2026

TITLE PAGE

Problem Statement ID – SIH26120

Problem Statement Title – Digital Twin for Well-to-Surface Optimization of Cyclic Steam Stimulation (CSS) and Sucker Rod Pump (SRP) Operations for Heavy Oil Wells of Baghewala Field

Organisation – Oil India Limited

Theme – Smart Automation

PS Category – Software

Team ID – <Team ID>

Team Name – <Team Name>



ThermoLift

AI Well-to-Surface Digital Twin for Baghewala

1,202 bbl/d
field output (Apr 2026)

33 of 52
wells producing

10–13k cP
crude viscosity at 50 °C

1,150 m
average reservoir depth

Sources: Oil India Ltd; Business Standard, Apr 2026



THERMOLIFT

PS 26120

More oil per cycle. Less steam. No rod float.

Proposed Solution / Approach

The gap: crude is 10,000–13,000 cP at 50 °C; after steam it thickens ~1000× but SPM stays fixed, causing rod float, fluid pound and high SOR.

Reservoir + Wellbore Twin: daily temperature, viscosity and rate forecast (P10–P90) and the heat profile up the 1,150 m tubing.

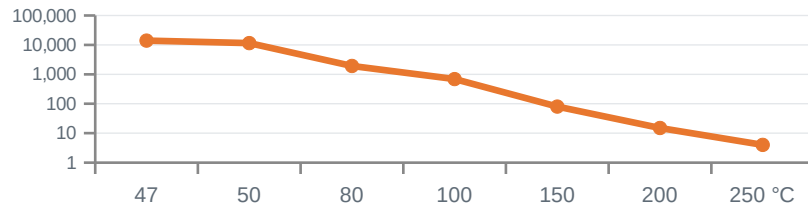
CSS Optimiser + Steam Scheduler: best steam, pressure, soak per cycle; plans mobile steam units across wells.

Adaptive SRP Control: rod-fall limit from rod weight vs drag of oil in the tubing; sets SPM, stroke and VFD downstroke.

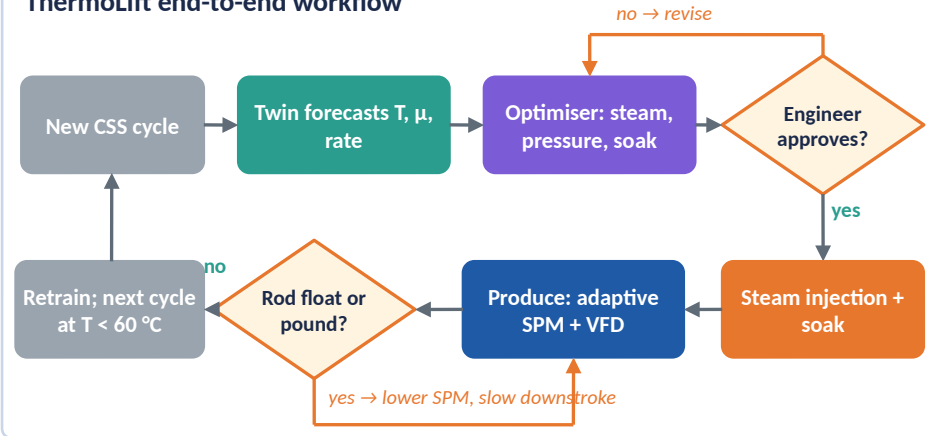
Pump-Health AI: surface and pump cards from the Gibbs wave equation, card diagnosis, rod-load and failure risk, heater cue.

Explainable Advisor: a reason for every setpoint; the engineer approves before any change.

Root cause: viscosity (cP, log) vs temperature



ThermoLift end-to-end workflow



Innovation and Uniqueness



Joint CSS + SRP loop
Reservoir, wellbore and pump optimised together



Viscosity-aware SPM
Pump speed follows the cooling well



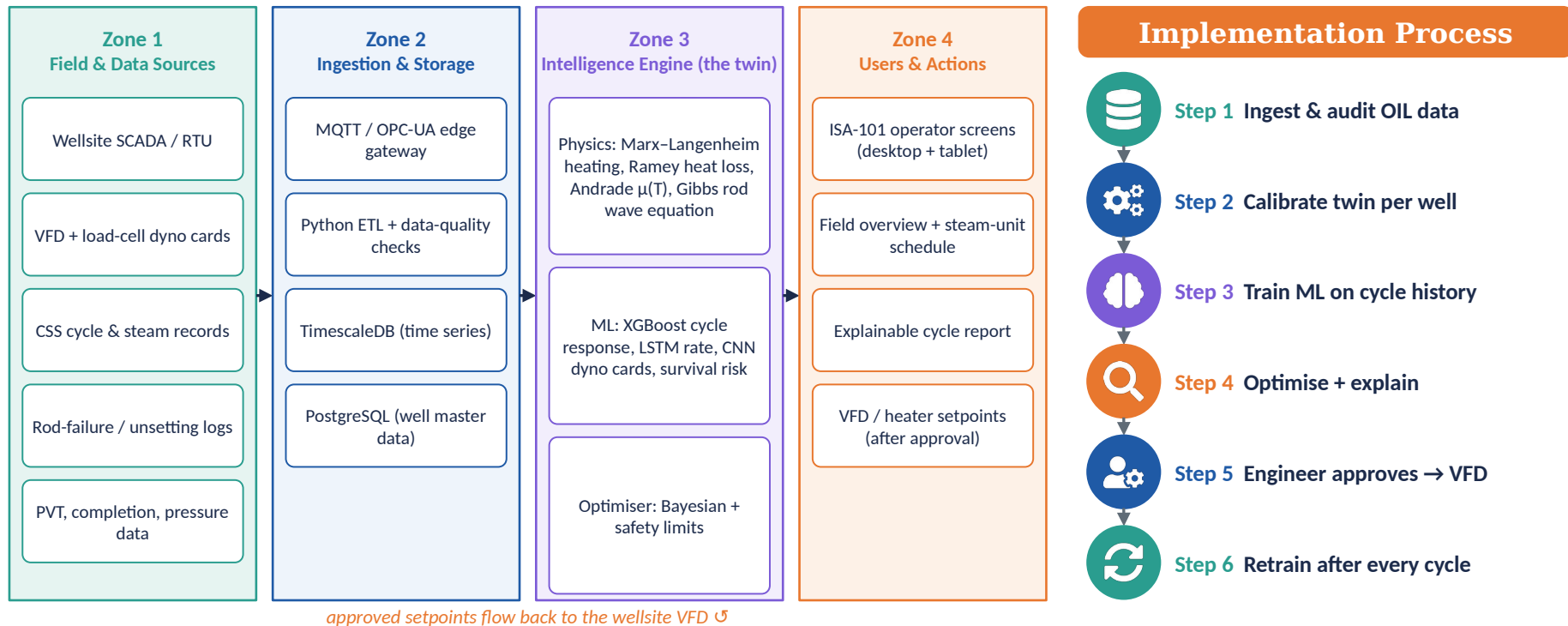
Physics + ML hybrid
Reliable with limited field data



Explainable AI
A reason for every recommendation



Operator-grade UX
ISA-101 screens; alarms by priority; a reason for every setpoint



Technology stack

Python

FastAPI

XGBoost

PyTorch

TimescaleDB

MQTT

React

D3.js

Docker

[Live demo ↗](#)
gauravchahal20.github.io/120-SIH

Feasibility

- Working prototype: reservoir-tubing-rod physics, optimiser, scheduler, 2 ML models; 17/17 automated checks pass
- Viscosity fitted to OIL's 10–13k cP at 50 °C
- Needs only data OIL already logs

Viability

- Cost $\approx$ ₹2 Cr (team, 33 edge gateways, servers)
- Benefit ₹12–48 Cr/yr: payback < 2 months even at 5% uplift
- Reuses existing VFDs, load cells, heaters

Sustainability

- Owned by OIL on-premise; open-source stack, no licence fees
- Retrains itself every cycle; run cost $\approx$ ₹30 L/yr
- Keeps expert know-how as data; lower SOR, lower CO₂

Implementation Plan

- Month 1: data audit, per-well calibration
- Months 2–3: advisory pilot on 3–5 CSS wells
- Months 4–6: field-wide rollout, then closed loop

Potential Challenges & Risks

01

Data gaps / noise: SCADA and cycle records may be incomplete.

02

Model drift: Reservoir response changes from cycle to cycle.

03

Unsafe setpoints: A wrong SPM could overload the rods.

04

OT cyber-security: SCADA links to wells are an attack surface.

05

User adoption: Operators trust experience over software.

Physics-informed models need less data; automatic quality checks and gap filling.

Automatic retraining after every completed CSS cycle.

Hard rod-load and SPM limits, and an engineer approves every change.

IEC 62443 zoning, read-only OPC-UA via edge gateway, no remote write without signed engineer approval.

Explainable reasons, advisory mode first, and pilot results shown on real wells.

01

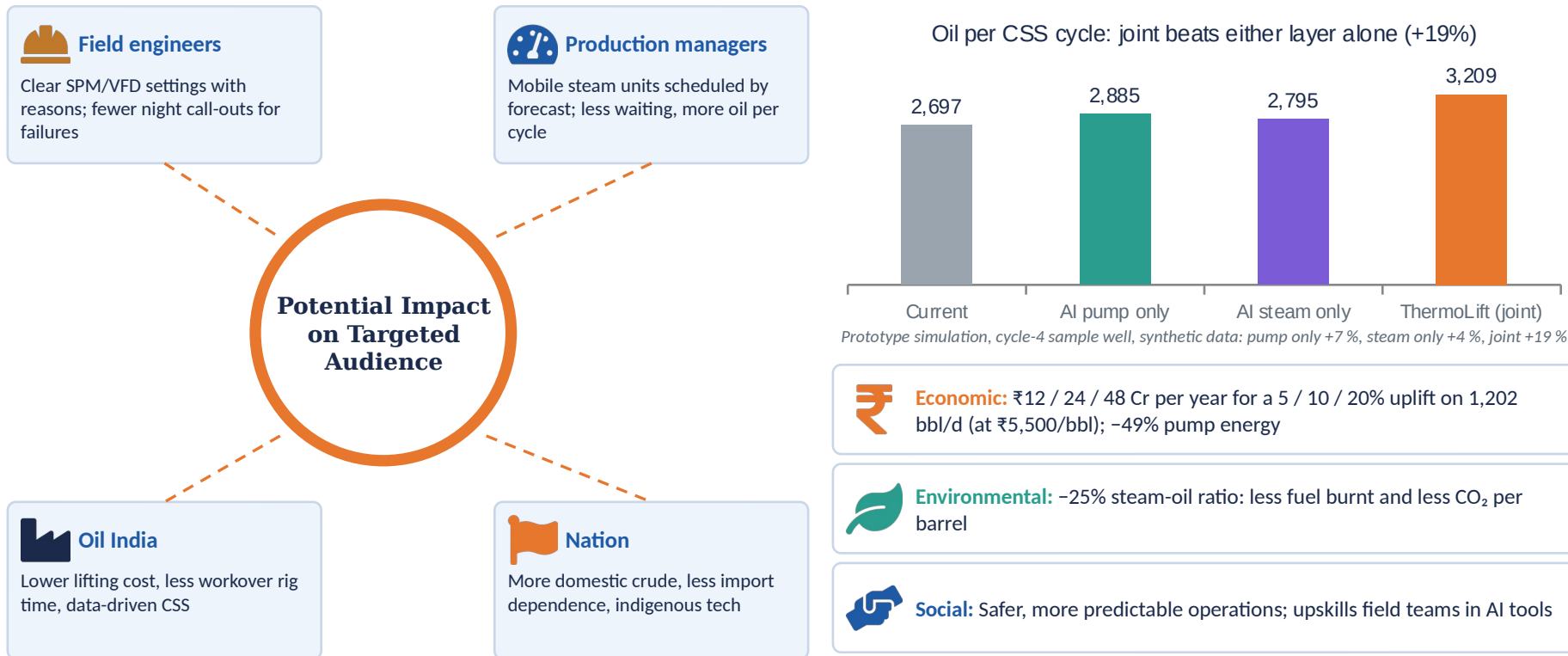
02

03

04

05

Strategies for Overcoming



Field & Applied Research

- Studied OIL's published Bagheewala data (depth, viscosity, CSS history, SRP lift) and calibrated the viscosity-temperature model to it.
- Built and verified a working prototype: 17/17 automated checks pass (physics, consistency, optimiser, ML). Cycle model R^2 0.80; pump-card classifier 94% on unseen cards.
- Honest scope: models are trained on physics-simulated data today; step 1 with OIL is retraining on real CSS, SCADA and failure records (CSV loader built in).

Sources

Oil India Ltd – Rajasthan Fields (Bagheewala): depth 1,150 m; 10,000–13,000 cP at 50 °C; CSS + SRP

Business Standard, Apr 2026 – OIL ramps up Rajasthan output: 1,202 bbl/d; CSS in 19 wells; downhole heaters

SPE 2024 – Cyclic steam injection for heavy oil in India and the Orinoco Belt

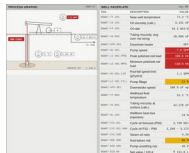
SRP design to avoid pump floating in heavy-oil, low-API wells

Models: Marx & Langenheim (1959); Ramey (1962) JPT; Butler (1991); Gibbs (2012)

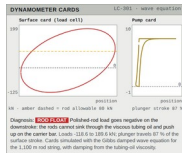
Future Scope & Business Potential

- Extend to OIL's planned SAGD and to other heavy-oil SRP fields, e.g. ONGC's Mehsana asset
- Add ESP / PCP lift models, real downhole cards and a field-crew mobile app
- Indigenous, licensable product for Indian E&Ps (Atmanirbhar Bharat)

Prototype evidence (live demo with guided tour, link on slide 3)



Process graphic + faceplate



Wave-equation cards: rod float



17/17 validation checks pass



Steam-unit schedule (1 unit)